



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

HALF YEARLY EXAMINATION

ASSESSMENT TASK 3

Year 11

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- Show all necessary working in Questions 11 – 14
- Write your Student Name **and** Class Teacher on the writing booklet(s) **or** sheet(s) submitted
- Weighting 30%

Total marks – 70

Section I

Pages 1 - 2

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 3 - 6

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

Section I**10 marks**

Shade the correct response on the answer sheet provided
Each question is worth 1 mark

1. What is the product of (6×10^3) , (4.6×10^5) and (2×10^{-2}) expressed in scientific notation?

A. 55.2×10^6 B. 5.52×10^7 C. 55.2×10^7 D. 5.52×10^{10}

2. The two lines represented by the equations below are graphed on a coordinate plane.

$$x + 6y = 12$$

$$3(x - 2) = -y - 4$$

Which statement best describes the two lines?

- A. The lines are parallel.
B. The lines are the same.
C. The lines are perpendicular.
D. The lines intersect at an angle other than 90 degrees.

3. The expression $\frac{\sin^2 \theta + \cos^2 \theta}{1 - \sin^2 \theta}$ is equivalent to

A. $\cos^2 \theta$ B. $\sin^2 \theta$ C. $\sec^2 \theta$ D. $\operatorname{cosec}^2 \theta$

4. When $x^{-1} - 1$ is divided by $x - 1$, the solution is:

A. -1 B. $-\frac{1}{x}$ C. $\frac{1}{x^2}$ D. $\frac{1}{(x-1)^2}$

5. Which ordered pair is in the solution set of the following system of linear equations?

$$y < 2x + 2$$

$$y \geq -x - 1$$

A. $(0, 3)$ B. $(2, 0)$ C. $(-1, 0)$ D. $(-1, -4)$

6. The solution set of the inequality $x^2 - 3x > 10$ is

- A. $-2 < x < 5$
- B. $0 < x < 3$
- C. $x < -5$ or $x > 2$
- D. $x < -2$ or $x > 5$

7. If $\frac{ey}{n} + k = t$, rewrite y as the subject of the equation.

- A. $y = \frac{tn+k}{e}$
- B. $y = \frac{tn-k}{e}$
- C. $y = \frac{n(t+k)}{e}$
- D. $y = \frac{n(t-k)}{e}$

8. The solution set of the equation $\sqrt{x+3} = 3-x$ is

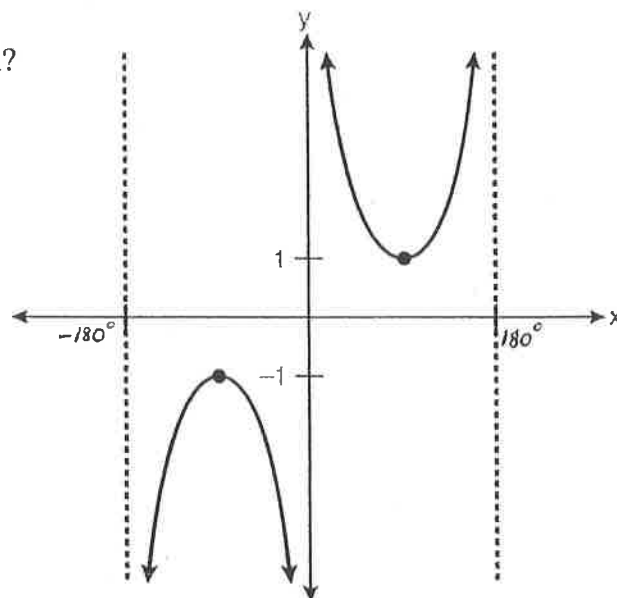
- A. $\{1\}$
- B. $\{0\}$
- C. $\{1, 6\}$
- D. $\{2, 3\}$

9. If $\triangle JKL \equiv \triangle MNO$, which statement is always true?

- A. $\angle KLJ = \angle NMO$
- B. $\angle KJL = \angle MON$
- C. $JL = MO$
- D. $JK = ON$

10. Which equation is sketched in the diagram?

- A. $y = \sec x$
- B. $y = \operatorname{cosec} x$
- C. $y = \tan x$
- D. $y = \cot x$



End of Section I

Section II**60 marks**

Begin each question on a new answer sheet

Show all necessary working

Each question is worth 15 marks

Question 11**(15 marks)****Begin a NEW answer booklet or answer sheet**

- (a) Simplify $(-2\sqrt{3})^3 \times (\sqrt{6})$ 1
- (b) Find the exact value of $\cos(-210^\circ)$. 1
- (c) Find $\lim_{x \rightarrow a} \frac{x^3 - a^3}{x^2 - a^2}$ 2
- (d) Solve $\frac{t+3}{5} - \frac{t-4}{3} = 2$ 2
- (e) i) State the domain of the function $y = \sqrt{9 - x^2}$. 1
ii) State the range of the function $y = 4 - |x|$. 1
- (f) Solve $4^{2x-5} = 8^{3x}$ 2
- (g) Show that $(k-1)x^2 + 2kx + k+1 = 0$ has real roots for all values of k . 2
- (h) If $x = \sqrt{2} - 1$, simplify $x^2 - \frac{1}{x^2}$ giving your answer with a rational denominator. 3

Question 12**(15 marks)****Begin a NEW answer booklet or answer sheet**

- (a) Sketch the graph of $y = (x + 2)^2 - 4$ showing the main features including the x and y intercepts. 2
- (b) The points A (1, 2), B (5, 4) and C (-3, 6) are located on a number plane.
- (i) What angle does the line AB make with the x - axis? 2
- (ii) Find the equation of the line parallel to AB passing through the point C. 2
- (iii) Find the perpendicular distance of the point C to the line AB. 2
- (iv) Find the area of triangle ABC 2
- (v) Find the circumference of the circle having BC as its diameter, correct to two decimal places. 2
- (c) Find the solutions of the following set of simultaneous equations. 3
- $$\begin{aligned}5 &= y - x \\ 4x^2 &= 4 + y - 17x\end{aligned}$$

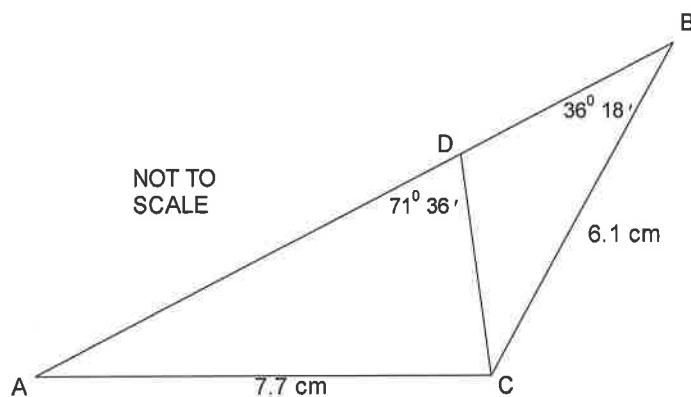
Question 13

(15 marks)

Begin a NEW answer booklet or answer sheet

- (a) Simplify $\frac{\operatorname{cosec} \theta}{\cos \theta + \sin \theta \tan \theta}$ 1

- (b) For the diagram below,



- (i) Find the length of CD. 2
- (ii) Find the length of AB. 2
- (c) Solve: $\sqrt{3} \tan 2\theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$ 3
- (d) If α and β are the roots of $3x^2 - 2x + 5 = 0$, find the value of $\alpha^2 + \beta^2$. 2
- (e) In the quadratic equation $x^2 + 3x + 2p - 1 = 0$, one root is triple another. Find the value of p . 2
- (f) Solve $\left(x + \frac{2}{x}\right)^2 + 4\left(x + \frac{2}{x}\right) + 3 = 0$ giving answer in exact form. 3

Question 14

(15 marks)

Begin a NEW answer booklet or answer sheet

(a) Differentiate $f(x) = 3x^2 - 4x + 2$ from first principles. 2

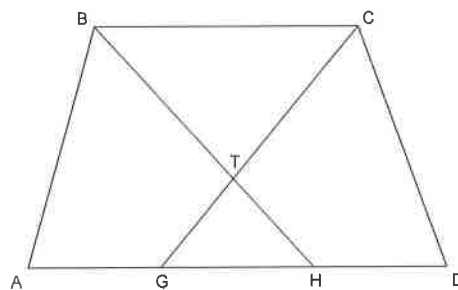
(b) Differentiate $y = \sqrt{8x} - \frac{1}{x}$ with respect to x . 2

(c) Differentiate $y = 3x^2(2x + 7)^3$ with respect to x . 2

(d) Find the equation of the tangent to the curve $y = \frac{x^2}{2x + 1}$ at the point where $x = -2$. 3

(e) Sketch the function $f(x) = \begin{cases} -4, & \text{if } x < 0 \\ x^2 - 4, & \text{if } 0 \leq x \leq 2 \\ x - 2, & \text{if } x > 2 \end{cases}$ 3

(f)



In the diagram, $\angle ABH = \angle DCG$, $BT = CT$, $TG = TH$.

Copy the diagram on to your answer sheet.

Prove: $AH = DG$ 3

End of Section II

End of Examination



TRINITY GRAMMAR SCHOOL
2012, Year 11 Mathematics Extension 1
Half Yearly Examination
SECTION I
ANSWER SHEET

SOLUTIONS

Name:

Class Teacher:

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

- | | | | | |
|------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Q1. | A | ☒ B | C | D |
| Q2. | A | B | C | ☒ D |
| Q3. | A | B | ☒ C | D |
| Q4. | A | ☒ B | C | D |
| Q5. | A | ☒ B | C | D |
| Q6. | A | B | C | ☒ D |
| Q7. | A | B | C | ☒ D |
| Q8. | ☒ A | B | C | D |
| Q9. | A | B | ☒ C | D |
| Q10. | A | ☒ B | C | D |

Q 11. a) $-8 \times 3\sqrt{3} \times \sqrt{6}$
 $= -24\sqrt{18}$
 $= -72\sqrt{2}$ (1)

b) $\cos(-210^\circ) = \cos 150^\circ$
 $= -\frac{\sqrt{3}}{2}$ (1)

c) $\lim_{x \rightarrow a} \frac{(x-a)(x^2+ax+a^2)}{(x-a)(x+a)}$ (M1)
 $= \frac{3a^2}{2a}$
 $= \frac{3a}{2}$ (1)

d) $3t + 9 - 5t + 20 = 30$ (M1)
 $-2t = 1$
 $t = -\frac{1}{2}$ (1)

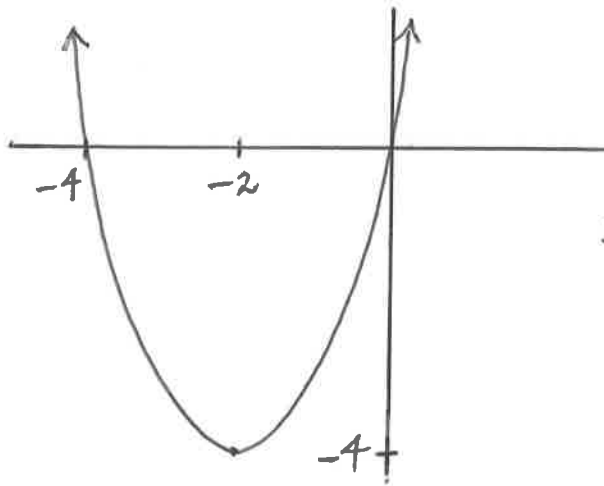
e) i) D: $-3 \leq x \leq 3$ (1)
 ii) R: $y \leq 4$ (1)

f) $2^{4x-10} = (2^3)^{3x}$
 $4x-10 = 9x$ (M1)
 $-10 = 5x$
 $x = -2$ (1)

g) $4k^2 - 4(k-1)(k+1) \geq 0$ (M1)
 $4k^2 - 4(k^2-1) \geq 0$
 $4k^2 - 4k^2 + 4 \geq 0$
 $4 \geq 0$ (1)
 since $\Delta \geq 0$ the roots are real.

h) $3-2\sqrt{2} - \frac{1}{(3-2\sqrt{2})} \times \frac{(3+2\sqrt{2})}{(3+2\sqrt{2})}$ (M1)
 $= 3-2\sqrt{2} - \frac{(3+2\sqrt{2})}{9-8}$ (M1)
 $= -4\sqrt{2}$ (1)

Q12. a)



x-intercepts - (1)
Vertex - (1)

b) i) $m_{AB} = \frac{2}{4} = \frac{1}{2}$ (1) $\theta = \tan^{-1}\left(\frac{1}{2}\right) = 26^\circ 34'$ (1)

ii) $y - 6 = \frac{1}{2}(x + 3)$ (M1)
 $2y - 12 = x + 3$
 $x - 2y + 15 = 0$ (1)

iii) Line AB: $y - 2 = \frac{1}{2}(x - 1)$
 $2y - 4 = x - 1$ (M1)
 $x - 2y + 3 = 0$

$d = \frac{|-3 - 2(6) + 3|}{\sqrt{1 + 4}} = \frac{12}{\sqrt{5}}$ (1)

iv) $AB = \sqrt{4^2 + 2^2}$
 $= \sqrt{20}$ (M1)

Area $= \frac{1}{2} \times 2\sqrt{5} \times \frac{12}{\sqrt{5}}$
 $= 12 \text{ m}^2$ (1)

v) $BC = \sqrt{2^2 + (-8)^2}$
 $= \sqrt{68}$ (M1)

$C = \pi \times \sqrt{68}$
 $= 25.91 \text{ units}$ (1)

c) $y = x + 5$
 $4x^2 = 4 + (x + 5) - 17x$
 $4x^2 + 16x - 9 = 0$ (M1)
 $(2x - 1)(2x + 9) = 0$
 $\begin{cases} x = \frac{1}{2} \\ y = 5\frac{1}{2} \end{cases}$ (1) $\begin{cases} x = -\frac{9}{2} \\ y = \frac{1}{2} \end{cases}$ (1)

$$Q13. a) \frac{\frac{1}{\sin \theta}}{\cos \theta + \frac{\sin^2 \theta}{\cos \theta}} = \frac{\frac{1}{\sin \theta}}{\frac{1}{\cos \theta}} = \cot \theta \quad (1)$$

$$b) i) \frac{CD}{\sin 36^\circ 18'} = \frac{6.1}{\sin 108^\circ 24'}$$

$$CD = \frac{6.1 \times \sin 36^\circ 18'}{\sin 108^\circ 24'} \quad (M1)$$

$$= 3.805850655 \quad (1)$$

$$ii) \frac{\sin A}{6.1} = \frac{\sin 36^\circ 18'}{7.7}$$

$$A = \sin^{-1} \left[\frac{6.1 \times \sin 36^\circ 18'}{7.7} \right]$$

$$= 27^\circ 58'$$

$$\therefore \angle ACB = 115^\circ 44' \quad (M1)$$

$$AB = \sqrt{7.7^2 + 6.1^2 - 2(7.7)(6.1) \cos 115^\circ 44'}$$

$$= 11.7 \quad (1)$$

$$c) \tan 2\theta = \frac{1}{\sqrt{3}}$$

$$2\theta = 30^\circ, 210^\circ, 390^\circ, 570^\circ \quad (M1)$$

$$\theta = 15^\circ, 105^\circ, 195^\circ, 285^\circ$$

$$d) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad (M1)$$

$$= \left(\frac{2}{3}\right)^2 - 2\left(\frac{5}{3}\right)$$

$$= -\frac{26}{9} \quad (1)$$

$$e) \alpha + 3\alpha = -3$$

$$4\alpha = -3$$

$$\alpha = -\frac{3}{4} \quad (M1)$$

$$3\alpha^2 = 2p - 1$$

$$\frac{27}{16} = 2p - 1$$

$$p = \frac{43}{32} \quad (1)$$

$$f) w = x + \frac{2}{x}$$

$$w^2 + 4w + 3 = 0 \quad (M1)$$

$$(w+3)(w+1) = 0$$

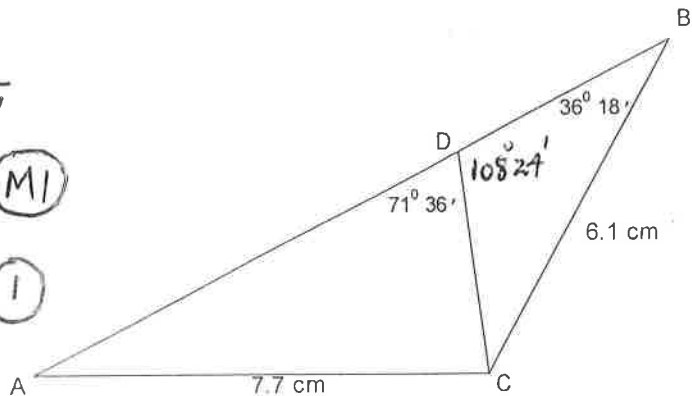
$$x + \frac{2}{x} + 3 = 0 \quad x + \frac{2}{x} + 1 = 0$$

$$x^2 + 3x + 2 = 0 \quad x^2 + x + 2 = 0$$

$$(x+2)(x+1) = 0 \quad \Delta = 1 - 4(1)(2)$$

$$x = -2 \quad x = -1 \quad = -7$$

$$(1) \quad \text{No real roots}$$



$$Q 14. a) f(x+h) = 3(x^2 + 2xh + h^2) - 4(x+h) + 2$$

$$= 3x^2 + 6xh + 3h^2 - 4x - 4h + 2$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 4x - 4h + 2 - 3x^2 + 4x - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 4)}{h}$$

$$= 6x - 4 \quad (1) \quad (M1)$$

$$b) y = \sqrt{8} x^{1/2} - x^{-1}$$

$$\frac{dy}{dx} = \sqrt{8} \cdot \frac{1}{2} x^{-1/2} + x^{-2}$$

$$= \sqrt{2} x^{-1/2} + \frac{1}{x^2} \quad (1)$$

$$c) y' = (3x^2) \cdot 3(2x+7)^2(2) + (2x+7)^3(6x) \quad (M1)$$

$$= 18x^2(2x+7)^2 + 6x(2x+7)^3$$

$$= 6x(2x+7)^2(3x + 2x+7)$$

$$= 6x(5x+7)(2x+7)^2 \quad (1)$$

$$d) y = \frac{x^2}{2x+1} \quad \text{when } x = -2 \quad y = \frac{4}{-3}$$

$$y' = \frac{(2x+1)(2x) - (x^2)(2)}{(2x+1)^2}$$

$$= \frac{2x^2 + 2x}{(2x+1)^2}$$

$$= \frac{2x(x+1)}{(2x+1)^2} \quad (M1)$$

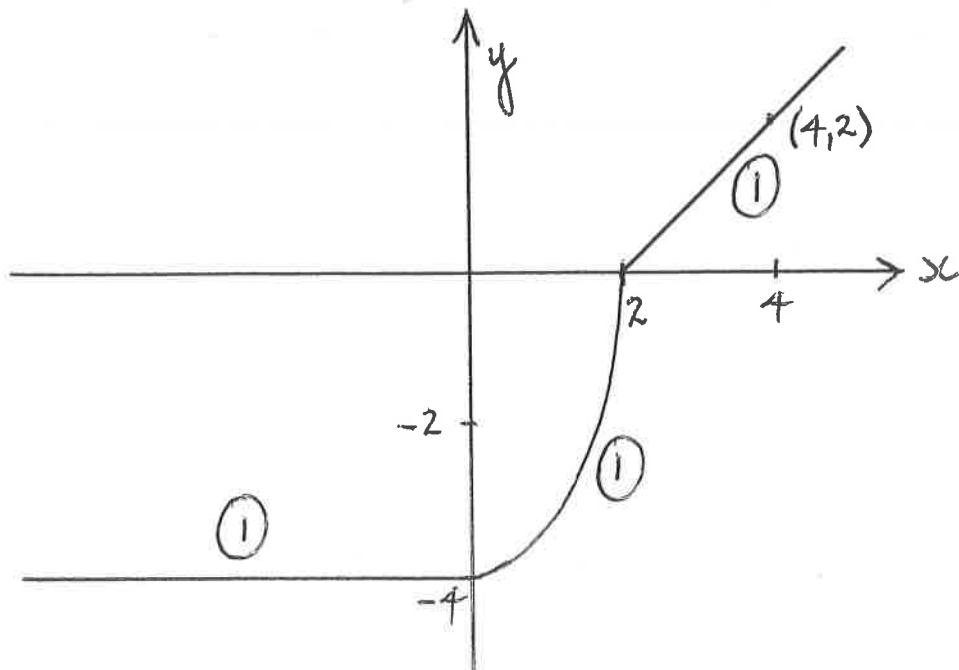
$$\text{when } x = -2 \quad y' = \frac{(-4)(-1)}{9} = \frac{4}{9}$$

$$y + \frac{4}{3} = \frac{4}{9}(x + 2) \quad (M1)$$

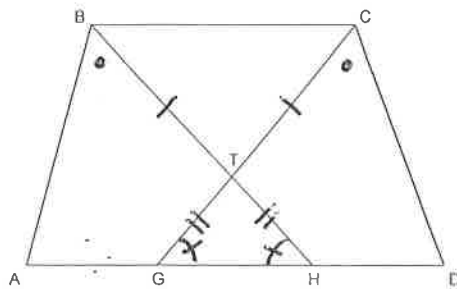
$$9y + 12 = 4x + 8$$

$$4x - 9y - 4 = 0 \quad (1)$$

Q14
e)



f)



$$\angle ABH = \angle DCG$$

Given

$$BT = CT$$

Given

$$TG = TH$$

Given

$$BH = CG$$

$$BT + TH = CT + TG$$

$$\angle TGH = \angle THG$$

Base L's in $\Delta =$ ①

$$\Delta ABH \equiv \Delta DCG$$

A.A.S ①

$$AH = DG$$

Matching sides $\equiv \Delta's =$ ①